

Problem 1 (Working with negligible functions). A non-negative function $\nu: \mathbb{N} \rightarrow \mathbb{R}$ is negligible if it decreases faster than the inverse of any polynomial; more precisely, for each polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficient), there exists some $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{P(n)}$ whenever $n > N$. Otherwise, we say that ν is non-negligible.

(i) Show that ν is negligible if and only if for every fixed sufficiently large integer c we have

$$\lim_{n \rightarrow \infty} \nu(n) \cdot n^c = 0.$$

(The displayed condition can be denoted by $\nu(n) = o(n^{-c})$.)

In the following, $\text{negl}(n)$ denotes some arbitrary negligible function, and $\text{poly}(n)$ denotes some arbitrary polynomial in n . State whether each of the following functions is negligible or non-negligible, and give a brief justification.

- (i) $\nu(n) = 1/2^{100 \log n}$.
- (ii) $\nu(n) = n^{-\log \log \log n}$.
- (iii) $\nu(n) = \text{poly}(n) \cdot \text{negl}(n)$. (State whether ν is always negligible, or not necessarily.)
- (iv) $\nu(n) = \text{negl}(n)^{\frac{1}{\text{poly}(n)}}$. (Same instructions as previous item.)
- (v) Prove or disprove the following statement: $\nu(n)$ is non-negligible if and only if there exists a polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficient), there exists some $N \in \mathbb{N}$ such that $\nu(n) \geq \frac{1}{P(n)}$ whenever $n > N$.
- (vi) Given finitely many tuples $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}$ show that there are always a negligible function ν and a non-negligible function f such that $\nu(n_i) = f(n_i) = y_i$ for each i . Does the same conclusion still hold for infinitely many such tuples?
- (vii) (I have no time to type in the description of this subproblem, sorry!)

(i). To prove the "if" part, suppose for every integer $c \geq N_0$ we have $\lim_{n \rightarrow \infty} \nu(n)n^c = 0$. Now let $P(n) = a_k n^k + \dots + a_1 n + a_0 \in \mathbb{R}[n]$ be an arbitrary polynomial with $k \in \mathbb{N}, a_k > 0$ (i.e. with non-negative leading coefficient) and define $c \triangleq \max(N_0, k)$. Note that for all

$$n > \left\lceil \max_{i=0, \dots, k-1} \sqrt[k-i]{\frac{k|a_i|}{a_k}} \right\rceil \in \mathbb{N}$$

(denoted by N_1), we have the inequalities

$$\frac{a_k}{k} n^k > |a_i| n^i, \quad i = 0, \dots, k-1, \quad (*)$$

which give us the following result:

$$P(n) = a_k n^k + \sum_{i=0, \dots, k-1} a_i n^i \geq a_k n^k - \sum_{i=0, \dots, k-1} |a_i| n^i = \sum_{i=0, \dots, k-1} \left(\frac{a_k}{k} n^k - |a_i| n^i \right) \stackrel{(*)}{>} 0. \quad (\clubsuit)$$

We can then observe that for all $n > N_1$, $P(n)$ is in fact bounded by the monomial $2a_k n^c$ since

$$\begin{aligned}
 P(n) &= a_k n^k + \sum_{i=0, \dots, k-1} a_i n^i \\
 &\leq a_k n^k + \sum_{i=0, \dots, k-1} |a_i| n^i \\
 &< a_k n^k + \sum_{i=0, \dots, k-1} \frac{a_k}{k} n^k && \text{using } (*) \\
 &= 2a_k n^k \leq 2a_k n^c,
 \end{aligned}$$

which implies

$$n^{-c} \cdot \frac{1}{2a_k} < \frac{1}{P(n)}. \quad (\diamond)$$

(Note also that in the above the properties that result in $(\diamond)$: $\forall n > N_1 (0 < P(n) < 2a_k n^c)$ are actually derived by assuming $k > 0$. This is because if $k = 0$, it's trivial that $\forall n > N_1 (0 < P(n) = a_k < 2a_k n^c)$.)

On the other hand, since $\lim_{n \rightarrow \infty} \nu(n) n^c = 0$ and $\frac{1}{2a_k} > 0$, there exists some $N_2 \in \mathbb{N}$ such that

$$\nu(n) n^c = |\nu(n) n^c - 0| < \frac{1}{2a_k} \quad (\circ)$$

whenever $n > N_2$. (Note that $\nu(n)$ is non-negative.)

Hence, for all $n > \max(N_1, N_2) \in \mathbb{N}$, we have the inequality

$$\nu(n) \stackrel{(\diamond)}{<} n^{-c} \cdot \frac{1}{2a_k} \stackrel{(\circ)}{<} \frac{1}{P(n)}.$$

This shows that $\nu(n)$ is negligible by definition.

To prove the "only if" part, we assume $\nu(n)$ is negligible and let $c \in \mathbb{N}$. For every $\varepsilon > 0$, we know there exists some $N_\varepsilon \in \mathbb{N}$ such that $\nu(n) < \frac{1}{\varepsilon^{-1} n^c}$ whenever $n > N_\varepsilon$ as $\nu(n)$ is negligible and $\varepsilon^{-1} n^c \in \mathbb{R}[n]$ is a polynomial with non-negative leading coefficient, whence we have

$$|\nu(n) n^c - 0| = \nu(n) n^c < \varepsilon \quad \text{for all } n > N_\varepsilon$$

where the first equality holds since $\nu(n)$ is non-negative.

Thus, by definition, $\lim_{n \rightarrow \infty} \nu(n) n^c = 0$.

■

(ii). Non-negligible.

This is because for any integer $c \geq 100$,

$$\lim_{n \rightarrow \infty} \nu(n) n^c = \lim_{n \rightarrow \infty} \frac{n^c}{2^{100 \log n}} = \lim_{n \rightarrow \infty} \frac{n^c}{n^{100}} = 1 \text{ or } \infty,$$

implying $\nu(n) = \Omega(n^{-c})$. This implies $\nu(n)$ is non-negligible since we've shown in (i) that $\nu(n)$ is negligible iff $\nu(n) = o(n^{-c})$ for any sufficiently large integer c . ■

(iii). Negligible.

This is because for any $c \in \mathbb{N}$ and $\varepsilon_c > 0$,

$$\begin{aligned} |\nu(n)n^c - 0| &= \nu(n)n^c = n^{c - \log \log \log n} \\ &< n^{c - (c+1)} && \text{since } n > 2^{2^{c+1}} \\ &= n^{-1} \\ &< \varepsilon_c && \text{since } n > \varepsilon_c^{-1} \end{aligned}$$

whenever $n > \left\lceil \max(\varepsilon_c^{-1}, 2^{2^{c+1}}) \right\rceil \in \mathbb{N}$, which, by definition, leads to $\lim_{n \rightarrow \infty} \nu(n)n^c = 0$. This implies $\nu(n)$ is negligible since we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n \rightarrow \infty} \nu(n)n^c = 0$ for any sufficiently large integer c . ■

(iv). Negligible.

To prove the negligibility, we resort to the basic definition of it this time. Let $\nu(n) \triangleq R(n)\mu(n)$ where $R(n) \in \mathbb{R}[n]$ is a polynomial with non-negative leading coefficient and $\mu(n)$ be a negligible function. For each polynomial $P(n) \in \mathbb{R}[n]$ with non-negative leading coefficient, since $\mu(n)$ is negligible and $Q(n) \triangleq P(n)R(n) \in \mathbb{R}[n]$ is also a polynomial with non-negative leading coefficient, by definition there exists some $N \in \mathbb{N}$ such that $\mu(n) < \frac{1}{Q(n)}$ whenever $n > N$. It follows that for all $n > N$,

$$\nu(n) = R(n)\mu(n) < R(n) \cdot \frac{1}{Q(n)} = R(n) \cdot \frac{1}{P(n)R(n)} = \frac{1}{P(n)}.$$

This, by definition, implies $\nu(n)$ is also negligible. ■

(v). Not necessarily.

If $\nu(n) \triangleq (2^{-n^2})^{\frac{1}{n}} = 2^{-n}$, then $\nu(n)$ IS negligible since $\lim_{n \rightarrow \infty} \nu(n)n^c = \lim_{n \rightarrow \infty} \frac{n^c}{2^n} = 0$ for any integer $c \in \mathbb{N}$ and we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n \rightarrow \infty} \nu(n)n^c = 0$ for any sufficiently large integer c .

However, if $\nu(n) \triangleq (2^{-n^2})^{\frac{1}{n^2}} = \frac{1}{2}$, then $\nu(n)$ is NON-negligible since $\lim_{n \rightarrow \infty} \nu(n)n^c = \lim_{n \rightarrow \infty} \frac{n^c}{2} = \frac{1}{2}$ or $\infty \neq 0$ for all integer $c \in \mathbb{N}$ and we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n \rightarrow \infty} \nu(n)n^c = 0$ for any sufficiently large integer c . ■

(vi). A counterexample that violates the "only if" part: $\nu(n) \triangleq (-1)^n + 1$.

For the polynomial $P(n) \triangleq n$, it's clear that $\nu(n) = 2 \not< \frac{1}{n} = \frac{1}{P(n)}$ for all even and positive n . Since there are infinitely many such n , there is no $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{P(n)}$ whenever $n > N$. Thus $\nu(n)$ is indeed non-negligible.

On the other hand, for any polynomial $Q(n) \in \mathbb{R}[n]$ with non-negative leading coefficient, WLOG, suppose that $Q(x) = a_k n^k + \dots + a_1 n + a_0$ and $k \in \mathbb{N}, a_k > 0$, and then it follows from the reasoning for (♣) that for all

$$n > \left\lceil \max_{i=0, \dots, k-1} \sqrt[k-i]{\frac{k|a_i|}{a_k}} \right\rceil \in \mathbb{N}$$

(denoted by N'), we have

$$Q(n) \stackrel{(\clubsuit)}{>} 0.$$

Now observe that for all odd $n > N'$,

$$\nu(n) = (-1)^n + 1 = 0 < \frac{1}{Q(n)}.$$

Since there are infinitely many such n , there is no $N \in \mathbb{N}$ such that $\nu(n) \geq \frac{1}{Q(n)}$ whenever $n > N$. Hence the LHS holds while RHS does not. ■

(vii). Let ν' and f' be an arbitrary negligible function and an arbitrary non-negligible function, respectively (say, for instance, $\nu'(n) = 2^{-n}$ and $f'(n) = n^{-1}$).

For finitely many tuples, say $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}, i = 0, 1, \dots, k-1$, we simply define

$$\nu(n) \triangleq \begin{cases} y_i & n = n_i \text{ for some } i \\ \nu'(n) & \text{otherwise} \end{cases}, \quad f(n) \triangleq \begin{cases} y_i & n = n_i \text{ for some } i \\ f'(n) & \text{otherwise} \end{cases}.$$

Observe that $\nu(n) = f(n) = y_i$ for each i whereas ν and f are still negligible and non-negligible, respectively, just as ν' and f' . The reason is that the definition of negligibility only discusses the behavior of functions when n is sufficiently large. Precisely speaking, one can derive the negligibility of ν by replacing N by $\max(N, \max_{i=0, \dots, k-1} n_i)$ in the definition of the negligibility of ν' ; likewise, the non-negligibility of f' means that there is a polynomial $P(n)$ with coefficients in $\mathbb{R}$ and non-negative leading coefficient such that $f'(n) \geq \frac{1}{P(n)}$ at infinitely many integers n , which must also hold for f and thus f is also non-negligible.

However, this is not the case anymore when there are infinitely many tuples, say $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}, i \in I$ where I is an infinite index set. Suppose $y_i = 1$ for every i and let ν be a negligible function. By definition, there is some $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{n}$ whenever $n > N$. This implies that $\nu(n_i) < \frac{1}{n_i} \leq 1 = y_i$ for any positive n_i ($i \in I$). Since n_i 's are distinct non-negative integers and I is an infinite set, there must be some $i \in I$ such that $n_i > 0$ and thus $\nu(n_i) \neq y_i$, which means the aforementioned conclusion does not hold anymore in this case. ■

Problem 2 (Large deviation bounds). Assume that $X_1, \dots, X_n$ are independent identically distributed (i.i.d.) random variables, each taking 1 with probability p and 0 with probability $1-p$. The Chernoff's bound says that for all $\varepsilon > 0$,

$$\Pr \left[\left| \frac{1}{n} \sum_i X_i - p \right| > \varepsilon \right] \leq 2e^{-n\varepsilon^2}.$$

If you are rusty on Chernoff's bound, read about it, e.g., [here](#) or search Google; there are lots of forms of the bound, the above being the most convenient for our applications.

- (i) How large should n be if we want the average of the X_i to be within $\pm\varepsilon$ of p with probability at least $1-\delta$? (asymptotic expression for n is enough)
- (ii) Imagine we used Chebyshev's bound instead of Chernoff's, and if you wish, assume for simplicity that $p = 1/2$. What bound on n would you get then? Do you see any advantage of Chebyshev's bound over Chernoff's?

(i). Observe that

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| \leq \varepsilon \right] = 1 - \Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| > \varepsilon \right] \geq 1 - 2e^{-n\varepsilon^2},$$

where the inequality is given by Chernoff's bound. Therefore, to arrive via Chernoff's bound at

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| \leq \varepsilon \right] \geq 1 - \delta,$$

the Chernoff's bound must be bounded below by $1 - \delta$ first, i.e., $1 - 2e^{-n\varepsilon^2} \geq 1 - \delta$, from which we deduce the requirement for n :

$$1 - 2e^{-n\varepsilon^2} \geq 1 - \delta \iff 2e^{-n\varepsilon^2} \leq \delta \iff n \geq -\varepsilon^{-2} \ln\left(\frac{\delta}{2}\right) = \varepsilon^{-2}(\ln(\delta^{-1}) + \ln 2) = \Theta(\varepsilon^{-2} \ln(\delta^{-1})).$$

Hence n should be $O(\varepsilon^{-2} \ln(\delta^{-1}))$. ■

(ii). Since $X_1, \dots, X_n$ are independent and Bernoulli distributed with probability p , i.e., $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$, we calculate the following:

$$\begin{aligned} E[X_i] &= p, & \text{Var}[X_i] &= E[X_i^2] - E[X_i]^2 = p - p^2 = p(1-p), \\ E \left[\frac{1}{n} \sum_{i=1}^n X_i \right] &= \frac{1}{n} \sum_{i=1}^n E[X_i] = p & \text{Var} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}[X_i] = \frac{p(1-p)}{n}. \end{aligned}$$

Thus the Chebyshev's bound says that, for all $\varepsilon > 0$

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - E \left[\frac{1}{n} \sum_{i=1}^n X_i \right] \right| > \varepsilon \right] \leq \frac{\text{Var} \left[\frac{1}{n} \sum_{i=1}^n X_i \right]}{\varepsilon^2},$$

i.e.,

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| > \varepsilon \right] \leq \frac{p(1-p)}{n\varepsilon^2}.$$

Similar to the reasoning in the [previous problem](#), observe that

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| \leq \varepsilon \right] = 1 - \Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| > \varepsilon \right] \geq 1 - \frac{p(1-p)}{n\varepsilon^2},$$

where the inequality is given by Chebyshev's bound. Therefore, to arrive via Chebyshev's bound at

$$\Pr \left[\left| \frac{1}{n} \sum_{i=0}^n X_i - p \right| \leq \varepsilon \right] \geq 1 - \delta,$$

the Chebyshev's bound must be bounded below by $1 - \delta$ first, i.e., $1 - \frac{p(1-p)}{n\varepsilon^2} \geq 1 - \delta$, from which we deduce the requirement for n :

$$1 - \frac{p(1-p)}{n\varepsilon^2} \geq 1 - \delta \iff \frac{p(1-p)}{n\varepsilon^2} \leq \delta \iff n \geq p(1-p)\delta^{-1}\varepsilon^{-2} \stackrel{\text{if } p=\frac{1}{2}}{=} \frac{1}{4}(\delta^{-1}\varepsilon^{-2}).$$

Hence n should be no less than $p(1-p)\delta^{-1}\varepsilon^{-2}$, or $n = O(\delta^{-1}\varepsilon^{-2})$ if p is constant.

We see some advantages of Chernoff's bound over Chebyshev's here:

1. Chernoff's bound is tighter than Chebyshev's, as the former decreases exponentially, being much faster than the latter, which decreases polynomially, both as functions of n . Meanwhile, the lower bound for n deduced from Chernoff's bound grows much slower and is thus tighter than that deduced from Chebyshev's: the former gives $O(\varepsilon^{-2} \ln(\delta^{-1}))$ while the latter gives $O(\delta^{-1}\varepsilon^{-2})$ (supposing p is a constant).
2. Chernoff's lower bound for n is independent of p , which is great compared to Chebyshev's!

As to the advantage of Chebyshev's bound over Chernoff's, I think one is that Chebyshev's bound, though being looser, requires less assumptions compared to Chernoff's (e.g. Chernoff's requires that $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$) and thus more general, that is, it can be applied in more settings as long as the mean and variance of the random variable. ■

Problem 3 (Union bounds). Let $A, B_1, \dots, B_k$ be some arbitrary probabilistic events.

(i) Show that

$$\Pr[B_1 \cup B_2 \cup \dots \cup B_k] \leq \sum_{i=1}^k \Pr[B_i].$$

(ii) Show that

$$\Pr[A \cap (B_1 \cup B_2 \cup \dots \cup B_k)] \leq \sum_{i=1}^k \Pr[A \mid B_i].$$

(iii) Let F be the set of functions mapping $\{1, 2, \dots, n\}$ to $\{1, 2, \dots, m\}$, where $m \geq n^2$. Show that

$$\Pr_{f \leftarrow F} [f \text{ is injective}] \geq 1 - \frac{n^2}{2m}.$$

Specifically, we use $f \leftarrow F$ to denote that f is chosen uniformly at random from F and a function is injective if $f(x) \neq f(y)$ for any distinct x, y .

(i). Define $B_1^* \triangleq B_1$ and $B_i^* \triangleq B_i \setminus \bigcup_{j=1}^{i-1} B_j$ for each $i = 2, 3, \dots, k$. Notice that $B_i^* \cap B_j^* = \emptyset \forall i \neq j$ and that $B_i^* \subseteq B_i \forall i$.

Claim 3.1. $\bigcup_{i=1}^k B_i = \bigcup_{i=1}^k B_i^*$

Proof. Let $\omega \in \bigcup_{i=1}^k B_i$. Then there must be some $1 \leq n \leq k$ such that $\omega \in B_n$. By the well-ordering of $\mathbb{Z}$, choose the smallest n_0 such that $\omega \in B_{n_0}$. Equivalently,

$$\omega \notin B_i \text{ for all } i = 1, \dots, n_0 - 1 \quad \text{and} \quad \omega \in B_{n_0}.$$

Hence,

$$\omega \in B_{n_0} \setminus \bigcup_{i=1}^{n_0-1} B_i = B_{n_0}^* \subseteq \bigcup_{i=1}^k B_i^*,$$

which means $\bigcup_{i=1}^k B_i \subseteq \bigcup_{i=1}^k B_i^*$.

Conversely, let $\omega \in \bigcup_{i=1}^k B_i^*$. Then there must be some $1 \leq n \leq k$ such that $\omega \in B_n^*$. Thus

$$\omega \in B_n^* = B_n \setminus \bigcup_{i=1}^{n-1} B_i \subseteq B_n \subseteq \bigcup_{i=1}^k B_i,$$

which means $\bigcup_{i=1}^k B_i^* \subseteq \bigcup_{i=1}^k B_i$. ■

Hence,

$$\begin{aligned} \Pr[B_1 \cup B_2 \cup \dots \cup B_k] &= \Pr \left[\bigcup_{i=1}^k B_i \right] \\ &= \Pr \left[\bigcup_{i=1}^k B_i^* \right] && \text{by the claim} \\ &= \sum_{i=1}^k \Pr[B_i^*] && \text{since } B_i^* \cap B_j^* = \emptyset \forall i \neq j \\ &\leq \sum_{i=1}^k \Pr[B_i] && \text{since } B_i^* \subseteq B_i \implies \Pr[B_i^*] \leq \Pr[B_i] \forall i. \end{aligned}$$

■

(ii).

$$\begin{aligned}
 \Pr[A \cap (B_1 \cup B_2 \cup \dots \cup B_k)] &= \Pr[(A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_k)] \\
 &\leq \sum_{i=1}^k \Pr[A \cap B_i] && \text{using the result of (i)} \\
 &= \sum_{i=1}^k \Pr[B_i] \Pr[A | B_i] \\
 &\leq \sum_{i=1}^k \Pr[A | B_i] && \text{since } \Pr[B_i] \leq 1 \forall i
 \end{aligned}$$

■

(iii). Let F be the sample space and denote by E_{xy} the event in which $f(x) = f(y)$ for each $x, y \in \{1, 2, \dots, n\}$, that is,

$$E_{xy} \triangleq \{f \in F \mid f(x) = f(y)\}.$$

Note that since f is uniformly chosen from F , for any distinct $x, y \in \{1, 2, \dots, n\}$,

$$\Pr_{f \leftarrow F}[E_{xy}] = \frac{|E_{xy}|}{|F|} = \frac{m \cdot m^{n-2}}{m^n} = \frac{1}{m}. \quad (\spadesuit)$$

This leads us to an upper bound for $\Pr_{f \leftarrow F}[f \text{ is not injective}]$:

$$\begin{aligned}
 \Pr_{f \leftarrow F}[f \text{ is not injective}] &= \Pr_{f \leftarrow F}\left[\bigcup_{1 \leq x < y \leq n} E_{xy}\right] && \text{by the definition of injectivity} \\
 &\leq \sum_{1 \leq x < y \leq n} \Pr_{f \leftarrow F}[E_{xy}] && \text{using the result of (i)} \\
 &= \sum_{1 \leq x < y \leq n} \frac{1}{m} && \text{by } (\spadesuit) \\
 &= \frac{1}{m} \cdot \binom{n}{2} = \frac{n(n-1)}{2m} \leq \frac{n^2}{2m}. && (\star)
 \end{aligned}$$

Hence,

$$\Pr_{f \leftarrow F}[f \text{ is injective}] = 1 - \Pr_{f \leftarrow F}[f \text{ is not injective}] \stackrel{(\star)}{\geq} 1 - \frac{n^2}{2m}.$$

■

Problem 4 (One-Way Functions). We first review the concept of one-way functions which we've introduced in the lecture. A function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$ is one-way if it can be efficiently computed, but no efficient algorithm can invert it. More precisely, for any probabilistic polynomial time (PPT) algorithm Adv ,

$$\Pr_{x \xleftarrow{\$} \{0, 1\}^n} [f(\text{Adv}(1^n, f(x))) = f(x)] = \text{negl}(n)$$

In this problem, we focus on one-way function f that is length-preserving, i.e., for every $x \in \{0, 1\}^*$ it holds that $|f(x)| = |x|$. Note that $|$ denotes concatenation, e.g., $1011|0000 = 10110000$, $\oplus$ denotes bit-by-bit XOR, $\wedge$ denotes bit-by-bit AND, and $x_{[a:b]}$ denotes the substring of x from the a th bit to the b th bits.

For each of the following functions g , assuming f is a one-way function, prove that g is a one-way function, or provide a counter example to demonstrate that it is not always a one-way function. More specifically:

- In the case that g is one-way, you should show a reduction from inverting f to inverting g . In other word, you should show that if g is efficiently invertible, then f is also efficiently invertible.
- In the case that g is not always one-way, you should do the following. Assuming only that length-preserving one-way functions exist, you should construct a length-preserving one-way function f such that when instantiated with f , g is not a one-way function. That is, you should show a polynomial-time inverting algorithm for g .

For example, to prove that the construction of $g(x) = f(x_{[1:n/2]}|0^{n/2})$ is not always one-way, the argument goes as follows. Assume that h is an arbitrary length-preserving one-way function. Construct the function

$$f(x) = \begin{cases} 0^n & x_{[n/2+1:n]} = 0^{n/2} \\ h(x) & \text{otherwise.} \end{cases}$$

f is one-way assuming that h is one-way (you have to show this by a reduction). Furthermore, g constructed using this f is not one-way (you have to show this by exhibiting a polynomial-time attack). In general, your solutions should contain complete proofs that the counter examples are valid. In the following, assume $|x| = n$,

- (i) $g(x) = f(x_{[1:\frac{n}{2}]})|x_{[\frac{n}{2}+1:n]}$
- (ii) $g(x) = f(x)|f(f(x))$
- (iii) $g(x) = f(f(x))$ where f is a one-way permutation, i.e., f is one-way and bijective.
- (iv) $g(x) = f(x) \oplus x$
- (v) $g(x|x') = f(x \wedge x')$ for all $x, x' \in \{0, 1\}^n$. Here $g: \{0, 1\}^{2n} \rightarrow \{0, 1\}^n$.

For the following subproblems we will prove a trivial property first:

Lemma 4.1. Let $\nu, \mu: \mathbb{N} \rightarrow \mathbb{R}$ be non-negative functions such that there exists $N \in \mathbb{N}$ such that $\nu(n) \geq \mu(n)$ for all $n > N$. If μ is non-negligible, then ν is also non-negligible.

Proof of Lemma 4.1. We prove this by contradiction.

Suppose to the contrary that ν is negligible, which implies from the conclusion we have in problem

1(i) that for every fixed sufficiently large integer c we have

$$\lim_{n \rightarrow \infty} \nu(n)n^c = 0;$$

equivalently, for any $\varepsilon_c > 0$ there is $N_{\varepsilon_c} \in \mathbb{N}$ such that $\nu(n)n^c = |\nu(n)n^c - 0| < \varepsilon_c$ whenever $n > N_{\varepsilon_c}$. Therefore, for every fixed sufficiently large integer c and any $\varepsilon_c > 0$, we have

$$|\mu(n)n^c - 0| = \mu(n)n^c \leq \nu(n)n^c < \varepsilon_c$$

whenever $n > \max(N_{\varepsilon_c}, N) \in \mathbb{N}$. Equivalently,

$$\lim_{n \rightarrow \infty} \mu(n)n^c = 0.$$

for every fixed sufficiently large integer c . Hence μ must be negligible as well, which is contradictory to the antecedent. ■

Lemma 4.2. Let $\nu, \mu : \mathbb{N} \rightarrow \mathbb{R}$ be non-negative functions such that there exists $N \in \mathbb{N}$ such that $\nu(n) \geq \mu(n)$ for all $n > N$. If ν is negligible then so is μ .

Proof of Lemma 4.2. This is simply the contrapositive of lemma 4.1. ■

(i). We prove g is one-way via reduction.

Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

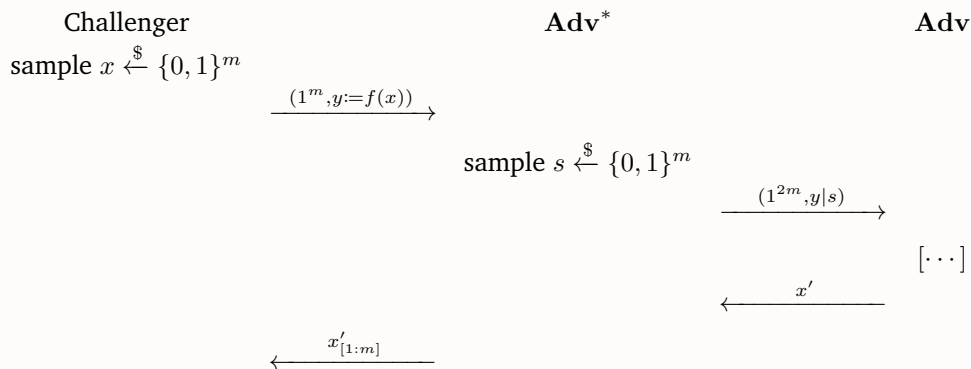
$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible. That is to say, $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g .) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0, 1\}^m$. Then, it uniformly samples $s \leftarrow \{0, 1\}^m$ and runs $\mathbf{Adv}$ on input $(1^{2m}, y|s)$, receiving as output x' . Finally, it outputs $x'_{[1:m]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{2m}, y|s)_{[1:m]} \text{ where } s \xleftarrow{\$} \{0, 1\}^m.$$

The following diagram illustrates the algorithm we've just constructed.



It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply samples m bits, concatenates bit strings once, runs $\mathbf{Adv}$ on a $2m$ -bit input, which also runs in PPT, and splits a bit string. For the second part, we claim that

$$\Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned} \Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x, s \leftarrow \{0,1\}^m} [f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]}) = f(x)] \\ &\geq \Pr_{x, s \leftarrow \{0,1\}^m} [g(\mathbf{Adv}(1^{2m}, f(x)|s)) = f(x)|s] \\ &= \Pr_{x^* \leftarrow \{0,1\}^{2m}} [g(\mathbf{Adv}(1^{2m}, f(x^*_{[1:m]})|x^*_{[m+1,2m]})) = f(x^*_{[1,m]})|x^*_{[m+1,2m]}] \\ &\quad (\text{since } x|s \text{ is distributed identically to a uniformly random } 2m\text{-bit string}) \\ &= \Pr_{x^* \leftarrow \{0,1\}^{2m}} [g(\mathbf{Adv}(1^{2m}, g(x^*))) = g(x^*)] \\ &\quad (\text{by def. of } g) \\ &= \varepsilon(2m), \end{aligned}$$

where the inequality comes from the fact that

$$\begin{aligned} g(\mathbf{Adv}(1^{2m}, f(x)|s)) = f(x)|s &\iff f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]})| \mathbf{Adv}(1^{2m}, f(x)|s)_{[m+1,2m]} = f(x)|s \\ &\implies f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]}) = f(x). \end{aligned}$$

As $\varepsilon(2m)$ is non-negligible, which follows from the fact that $\varepsilon(m)$ is non-negligible, by lemma 4.1,

$$\Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. ■

(ii). We prove g is one-way via reduction.

Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

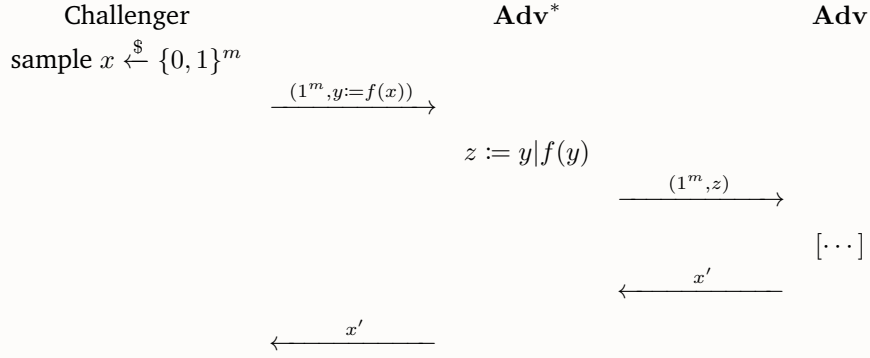
$$\varepsilon(n) \triangleq \Pr_{x \leftarrow \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible, i.e., that $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g .) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^m, y|f(y))$, receiving as output x' . Finally, it simply outputs x' . Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^m, y|f(y)).$$

The following diagram illustrates the algorithm we've just constructed.



It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply computes f , which is computable in PPT as f is one-way by assumption, concatenates bit strings once, and then runs $\mathbf{Adv}$ on a $2m$ -bit input, which also runs in PPT. For the second part, we claim that

$$\Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
 \Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)] \\
 &\stackrel{(a)}{=} \Pr_{x \leftarrow \{0,1\}^m} [g(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)|f(f(x))] \\
 &= \Pr_{x \leftarrow \{0,1\}^m} [g(\mathbf{Adv}(1^m, g(x))) = g(x)] \\
 &= \varepsilon(m),
 \end{aligned}$$

where equality (a) comes from the fact that

$$\begin{aligned}
 &g(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)|f(f(x)) \\
 \iff &f(\mathbf{Adv}(1^m, f(x)|f(f(x))))|f(f(\mathbf{Adv}(1^m, f(x)|f(f(x))))) = f(x)|f(f(x)) \\
 \iff &f(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x).
 \end{aligned}$$

As $\varepsilon(m)$ is non-negligible by assumption,

$$\Pr_{x \leftarrow \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. $\blacksquare$

(iii). We prove g is one-way via reduction.

Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

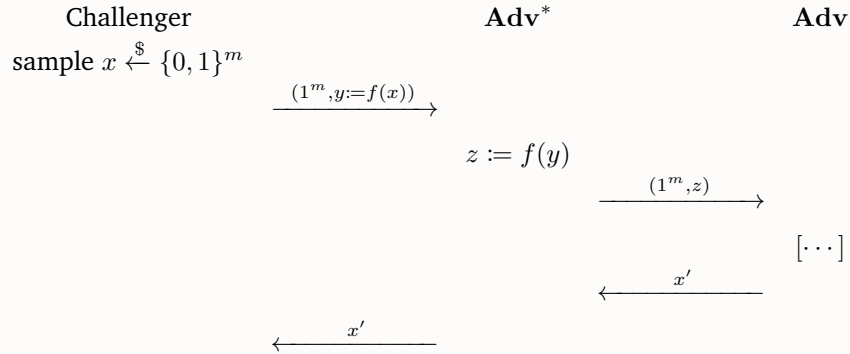
$$\varepsilon(n) \triangleq \Pr_{x \leftarrow \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible, i.e., that $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g .) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0, 1\}^m$. Then, it computes $z := f(y)$ and runs $\mathbf{Adv}$ on input $(1^m, z)$, receiving as output x' . Finally, it simply outputs x' . Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^m, f(y)).$$

The following diagram illustrates the algorithm we've just constructed.



It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply computes f , which is computable in PPT as f is one-way by assumption, and then runs $\mathbf{Adv}$ on an m -bit input, which also runs in PPT. For the second part, we claim that

$$\Pr_{x \leftarrow \{0, 1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
 \Pr_{x \leftarrow \{0, 1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x \leftarrow \{0, 1\}^m} [f(\mathbf{Adv}(1^m, f(f(x)))) = f(x)] \\
 &\stackrel{(a)}{=} \Pr_{x \leftarrow \{0, 1\}^m} [g(\mathbf{Adv}(1^m, f(f(x)))) = f(f(x))] \\
 &= \Pr_{x \leftarrow \{0, 1\}^m} [g(\mathbf{Adv}(1^m, g(x))) = g(x)] \\
 &= \varepsilon(m),
 \end{aligned}$$

where equality (a) comes from the fact that f is injective (as it is a permutation by assumption) and therefore

$$\begin{aligned}
 g(\mathbf{Adv}(1^m, f(f(x)))) &= f(f(x)) \stackrel{\text{definition of } g}{\iff} f(f(\mathbf{Adv}(1^m, f(f(x)))) = f(f(x)) \\
 &\stackrel{\text{injectivity of } f}{\iff} f(\mathbf{Adv}(1^m, f(f(x)))) = f(x).
 \end{aligned}$$

As $\varepsilon(m)$ is non-negligible by assumption,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. ■

(iv). We show g is not always one-way by providing a counterexample, i.e., a length-preserving one-way function f such that the corresponding g is not one-way.

Let h be an arbitrary length-preserving one-way function (which exists by the assumption that a length-preserving one-way function f exists). We define our $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ as

$$f(x) \triangleq h(x_{[\frac{n}{2}+1, n]}) || 0^{\frac{n}{2}} \text{ where } n \text{ denotes } |x|,$$

which we claim to be length-preserving and one-way.

Claim 4.1. f is a length-preserving one-way function.

Proof. Firstly, it's immediate that f is length-preserving as h is as well. Notice that f can be computed in PPT since h is one-way and thus can be computed in PPT by definition. Then, we prove f is one-way via reduction.

Suppose to the contrary that f is not one-way. As f can be computed in PPT, it must be that there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [f(\mathbf{Adv}(1^n, f(x))) = f(x)]$$

is non-negligible. Then, we construct another PPT algorithm $\mathbf{Adv}^*$ that inverts h with non-negligible probability. Since h is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := h(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^{2m}, y|0^m)$, receiving as output x' , and finally outputs $x'_{[m+1,2m]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{2m}, y|0^m)_{[m+1,2m]}.$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts h with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply concatenates bit strings, runs $\mathbf{Adv}$ on a $2m$ -bit input, which also runs in PPT, and then splits a bit string. For the second part, we claim

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned} \Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]}) = h(x)] \\ &\stackrel{(a)}{=} \Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}(1^{2m}, h(x)|0^m)) = h(x)|0^m] \\ &= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} [f(\mathbf{Adv}(1^{2m}, h(x^*_{[m+1,2m]}))|0^m) = h(x^*_{[m+1,2m]})|0^m] \\ &\quad (\text{since } x \text{ is distributed identically to the last } m \text{ bits of } x^* \xleftarrow{\$} \{0,1\}^{2m}) \\ &= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} [f(\mathbf{Adv}(1^{2m}, f(x^*))) = f(x^*)] = \varepsilon(2m), \end{aligned}$$

where equality (a) comes from the fact that

$$\begin{aligned} f(\mathbf{Adv}(1^{2m}, h(x)|0^m)) = h(x)|0^m &\iff h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]})|0^m = h(x)|0^m \\ &\iff h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]}) = h(x). \end{aligned}$$

As $\varepsilon(m)$ is non-negligible by assumption and so is $\varepsilon(2m)$,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts h with non-negligible probability, contradicting the assumption that h is one-way. $\blacksquare$

Next, to show g is not one-way given the f we've just defined, we provide a PPT algorithm $\mathbf{Adv}^\star$ that inverts g with non-negligible probability, i.e., that

$$\nu(n) \triangleq \Pr_{x \leftarrow \{0,1\}^n} \left[g(\mathbf{Adv}^\star(1^n, g(x))) = g(x) \right]$$

is non-negligible. We construct $\mathbf{Adv}^\star$ as follows: $\mathbf{Adv}^\star$ takes as input 1^n and $y := g(x)$, where $x \leftarrow \{0,1\}^n$. Then, it outputs $h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]}$. Namely,

$$\mathbf{Adv}^\star(1^n, y) \triangleq h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]}$$

It suffices to verify that $\mathbf{Adv}^\star$ is a PPT algorithm that inverts g with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^\star$ runs in PPT as it simply computes h , which is computable in PPT as it is one-way by assumption, taking XOR, and concatenates bit strings together. For the second part, observe that for any $n \in \mathbb{N}$

$$\begin{aligned} \nu(n) &\triangleq \Pr_{x \leftarrow \{0,1\}^n} \left[g(\mathbf{Adv}^\star(1^n, g(x))) = g(x) \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[g(h(g(x)_{[\frac{n}{2}+1, n]}) \oplus g(x)_{[1, \frac{n}{2}]}) = g(x) \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[g(h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]}) = y \right] \\ &\quad (\text{denote } g(x) \text{ by } y) \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[f \left(h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]} \right) \oplus \left(h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]} \right) = y \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[\left(h(y_{[\frac{n}{2}+1, n]}) \oplus 0^{\frac{n}{2}} \right) \oplus \left(h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]} \right) = y \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[\left(h(y_{[\frac{n}{2}+1, n]}) \oplus h(y_{[\frac{n}{2}+1, n]}) \oplus y_{[1, \frac{n}{2}]} \right) \mid \left(0^{\frac{n}{2}} \oplus y_{[\frac{n}{2}+1, n]} \right) = y \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} \left[y_{[1, \frac{n}{2}]} \mid y_{[\frac{n}{2}+1, n]} = y \right] \\ &= \Pr_{x \leftarrow \{0,1\}^n} [y = y] = 1. \end{aligned}$$

This implies $\nu(n)$ is non-negligible, following from the conclusion of problem 1(i) and the fact that

$$\lim_{n \rightarrow \infty} \nu(n) \cdot n^c = \lim_{n \rightarrow \infty} 1 \cdot n^c = 1 \text{ or } \infty \neq 0$$

for any fixed integer $c \in \mathbb{N}$. This proves that $\mathbf{Adv}^\star$ runs in PPT and inverts g with non-negligible probability, yielding that g is not one-way. ■

(v). We show g is not always one-way by providing a counterexample, i.e., a length-preserving one-way function f such that the corresponding g is not one-way.

Let h be an arbitrary length-preserving one-way function (which exists by the assumption that a length-preserving one-way function f exists) and c be an arbitrary real number in $\left(\frac{1}{4}, \frac{1}{2}\right)$ (say,

$c \triangleq \frac{1.14514}{4}$). We define our $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ as

$$f(x) \triangleq \begin{cases} 0^n & \text{if } \frac{|x|_1}{n} < c \\ 1|h(x_{[2,n]}) & \text{otherwise} \end{cases}$$

where we denote $n \triangleq |x|$ and define $| \cdot |_1$ as the number of 1's in a bit string, i.e.,

$$|b_1 b_2 \dots b_n|_1 \triangleq \sum_{i=1}^n b_i \quad \text{for each } x = b_1 b_2 \dots b_n \in \{0, 1\}^*.$$

We claim that f is length-preserving and one-way, with a detailed proof deferred until later (see claim 4.3). Next, we claim that g is not one-way given the f we've just defined, which is supposed to be length-preserving and one-way. Likewise, we postpone the detailed proof of g being non-one-way until later (see claim 4.4).

Before verifying the (non-)one-wayness of f and g , we make some observations that come in handy later. Notice that given an n -bit string $x = B_1 B_2 \dots B_n \xleftarrow{\$} \{0, 1\}^n$ sampled uniformly, each bit of x is in fact an independent, Bernoulli distributed random variable B_i with probability $p = \frac{1}{2}$, i.e., $B_1, B_2, \dots, B_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{2})$. Therefore the Chernoff's bound says that

$$\begin{aligned} \mu_1(n) &\triangleq \Pr_{x \xleftarrow{\$} \{0, 1\}^n} \left[\frac{|x|_1}{n} < c \right] = \Pr_{B_1, \dots, B_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{2})} \left[\frac{1}{n} \sum_{i=1}^n B_i < c \right] \\ &\leq \Pr_{B_1, \dots, B_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{2})} \left[\left| \frac{1}{n} \sum_{i=1}^n B_i - \frac{1}{2} \right| \geq \frac{1}{2} - c \right] \\ &\leq 2e^{-n(\frac{1}{2}-c)^2}. \end{aligned} \quad (\text{🙄})$$

Since $\lim_{n \rightarrow \infty} 2e^{-n(\frac{1}{2}-c)^2} n^{c'} = 0$ for any $c' \in \mathbb{N}$ and thus $2e^{-n(\frac{1}{2}-c)^2}$ is negligible (by problem 1(i)), by lemma 4.2 it follows that μ_1 is negligible.

Moreover, given 2 uniformly and independently sampled n -bit strings $x_1 = B_1 B_2 \dots B_n, x_2 = C_1 C_2 \dots C_n \xleftarrow{\$} \{0, 1\}^n$, we know that $B_1, \dots, B_n, C_1, \dots, C_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{2})$. Then each bit of the bitwise AND $x_1 \wedge x_2 = D_1, \dots, D_n$ of x_1 and x_2 is in fact an independent, Bernoulli distributed random variable D_i with probability $p = \frac{1}{4}$ since $\Pr[D_i = 1] = \Pr[B_i \wedge C_i = 1] = \Pr[B_i = 1] \Pr[C_i = 1] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; equivalently, $D_1, D_2, \dots, D_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{4})$. Likewise, the Chernoff's bound hence says that

$$\begin{aligned} \mu_2(n) &\triangleq \Pr_{x_1, x_2 \xleftarrow{\$} \{0, 1\}^n} \left[\frac{|x_1 \wedge x_2|_1}{n} < c \right] = \Pr_{D_1, \dots, D_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{4})} \left[\frac{1}{n} \sum_{i=1}^n D_i < c \right] \\ &\geq \Pr_{D_1, \dots, D_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{4})} \left[\left| \frac{1}{n} \sum_{i=1}^n D_i - \frac{1}{4} \right| < c - \frac{1}{4} \right] \\ &= 1 - \Pr_{D_1, \dots, D_n \stackrel{\text{iid}}{\sim} \text{Ber}(\frac{1}{4})} \left[\left| \frac{1}{n} \sum_{i=1}^n D_i - \frac{1}{4} \right| \geq c - \frac{1}{4} \right] \\ &\geq 1 - 2e^{-n(c-\frac{1}{4})^2}, \end{aligned} \quad (\text{🙄})$$

which is non-negligible as 1 is clearly non-negligible and $2e^{-n(c-\frac{1}{4})^2}$ has been shown to be negligible. Thus, from lemma 4.1 it follows that μ_2 is non-negligible.

Observation 4.2. μ_1 is negligible while μ_2 is non-negligible.

It remains to actually show that (1) f is length-preserving and one-way and that (2) g is not one-way given f .

Claim 4.3. f is a length-preserving one-way function.

Proof. Firstly, it's immediate that f is length-preserving as h is as well. Notice that f can be computed in PPT since h is one-way and thus can be computed in PPT by definition. Then, we prove f is one-way via reduction.

Suppose to the contrary that f is not one-way. As f can be computed in PPT, it must be that there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [f(\mathbf{Adv}(1^n, f(x))) = f(x)]$$

is non-negligible. Then, we construct another PPT algorithm $\mathbf{Adv}^*$ that inverts h with non-negligible probability. Since h is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := h(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^{m+1}, 1|y)$, receiving as output x' , and finally outputs $x'_{[2,m+1]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{m+1}, 1|y)_{[2,m+1]}.$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts h with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply concatenates bit strings, runs $\mathbf{Adv}$ on an $(m+1)$ -bit input, which also runs in PPT, and then splits a bit string. For the second part, we claim

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is non-negligible.

■

cont'd. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}(1^{m+1}, 1|h(x))_{[2,m+1]}) = h(x)] \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}} [h(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]})))_{[2,m+1]} = h(x^*_{[2,m+1]})] \\
&\stackrel{(a)}{\geq} \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}} \left[f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*) \wedge \frac{|x^*|_1}{m+1} \geq c \right] \\
&\stackrel{(b)}{\geq} \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}} [f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*)] - \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}} \left[\frac{|x^*|_1}{m+1} < c \right] \\
&= \varepsilon(m+1) - \mu_1(m+1),
\end{aligned}$$

where inequality (a) comes from the fact that

$$\begin{aligned}
f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*) \wedge \frac{|x^*|_1}{m+1} \geq c &\implies f(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]}))) = 1|h(x^*_{[2,m+1]}) \\
&\implies 1|h(\mathbf{Adv}(1^{m+1}, 1|h(x))_{[2,m+1]}) = 1|h(x^*_{[2,m+1]}) \\
&\iff h(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]})))_{[2,m+1]} = h(x^*_{[2,m+1]}),
\end{aligned}$$

and inequality (b) we have used the union bounds proved in problem 3(i) (by taking set complement on both sides).

Since $\varepsilon(m+1)$ is non-negligible and $\mu_1(m+1)$ is negligible by observation 4.2, $\varepsilon(m+1) - \mu_1(m+1)$ is non-negligible. Therefore,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts h with non-negligible probability, contradicting the assumption that h is one-way. $\blacksquare$

Claim 4.4. g is not one-way given the f defined above.

Proof. To see this, we provide a PPT algorithm $\mathbf{Adv}^\star$ that inverts g with non-negligible probability, i.e., that

$$\nu(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(\mathbf{Adv}^\star(1^{2n}, g(x))) = g(x) \right]$$

is non-negligible. We construct $\mathbf{Adv}^\star$ as follows: $\mathbf{Adv}^\star$ takes as input 1^{2n} and $y := g(x)$, where $x \leftarrow \{0,1\}^{2n}$. Then, it directly outputs (i.e. “guesses”) 0^{2n} . Namely,

$$\mathbf{Adv}^\star(1^n, y) \triangleq 0^{2n}.$$

It suffices to verify that $\mathbf{Adv}^\star$ is a PPT algorithm that inverts g with non-negligible probability. Indeed, by construction $\mathbf{Adv}^\star$ trivially runs in PPT. As for the second part, observe that for any $n \in \mathbb{N}$

$$\begin{aligned} \nu(n) &\triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(\mathbf{Adv}^\star(1^{2n}, g(x))) = g(x) \right] \\ &= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(0^{2n}) = g(x) \right] \\ &= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[f(0^n) = g(x) \right] \\ &= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[0^n = g(x) \right] \\ &= \Pr_{x_1, x_2 \xleftarrow{\$} \{0,1\}^n} \left[0^n = f(x_1 \wedge x_2) \right] \\ &= \Pr_{x_1, x_2 \xleftarrow{\$} \{0,1\}^n} \left[\frac{|x_1 \wedge x_2|_1}{n} < c \right] \\ &= \mu_2(n), \end{aligned}$$

which is non-negligible by observation 4.2. This proves that $\mathbf{Adv}^\star$ runs in PPT and inverts g with non-negligible probability, yielding that g is not one-way. ■

Now we’ve completed the proof. ■

Reference

P1 (None)

P2 • Mathematical statistics (MATH4013) lecture notes ([topic_2.pdf](#), [topic_3.pdf](#))

P3 • Mathematical statistics (MATH4013) lecture notes ([topic_1.pdf](#))

P4 • [Examples of reductions for one-way functions by Noah Stephens-Davidowitz](#)